

TRIGONOMETRIC RATIOS



SYNOPSIS

- **Angle:** An angle is the union of two rays having a common end point in a plane.
- **Measurement of an angle :**
- **Sexagesimal system:**
 - (i) One right angle = $\frac{\pi}{2}$ radian = 90° .
 - (ii) π radian = 2 right angles = 180° .
 - (iii) $1^\circ = 60$ minutes ($60'$)
 - (iv) $1' = 60$ seconds ($60''$)
 - (v) $1^\circ = 0.001745$ radian
 - (vi) $1 \text{ radian} \cong 57^\circ 17' 45''$ (approx)
- **Centesimal system:**
 - (i) 1 right angle = 100 grades written as 100^g
 - (ii) 1 grade or $1^g = 100$ minutes ($100'$)
 - (iii) 1 minute or $1' = 100$ seconds ($100''$)
- **Circular system :**
- **Radian:** A radian is the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle. The length of arc $l = r\theta$.
 - (i) 1 revolution = 2π radians = 360°
 - (ii) π radians = 2 right angles = $2 \times 90^\circ = 180^\circ$
 - (iii) 1 degree (1°) = $\frac{\pi}{180}$ rad $\cong 0.01745$ rad
 - (iv) 1 rad (1^c) = $\frac{180}{\pi}$ degrees $\cong 57^\circ 17' 46''$
- **Note:** (i) Value of $\pi = \frac{22}{7}$ (or) $\frac{355}{113}$ (or) 3.1416
 - (ii) π is an irrational number
 - (iii) $\pi = \frac{\text{circumference of the circle}}{\text{diameter of the circle}}$

Trigonometric Identities :

- i) $(\sin \theta)(\operatorname{cosec} \theta) = 1, \theta \neq n\pi, n \in Z$
- ii) $(\cos \theta)(\sec \theta) = 1, \theta \neq (2n+1)\frac{\pi}{2}, n \in Z$
- iii) $(\tan \theta)(\cot \theta) = 1, \theta \neq (2n+1)\frac{\pi}{2}, \theta \neq n\pi$
- iv) $\sin^2 \theta + \cos^2 \theta = 1 \Rightarrow \sin^2 \theta = 1 - \cos^2 \theta$
- v) $\sec^2 \theta - \tan^2 \theta = 1 \Rightarrow \sec^2 \theta = 1 + \tan^2 \theta$

$$\left(\theta \neq (2n+1)\frac{\pi}{2}, n \in Z \right).$$

$$\begin{aligned} \text{(vi) } \cos^2 \theta - \cot^2 \theta &= 1 \\ \Rightarrow \cos^2 \theta &= 1 + \cot^2 \theta \quad (\theta \neq n\pi, n \in Z) \end{aligned}$$

- **Note:** (i) If $\theta \neq (2n+1)\frac{\pi}{2}, n \in Z$, then

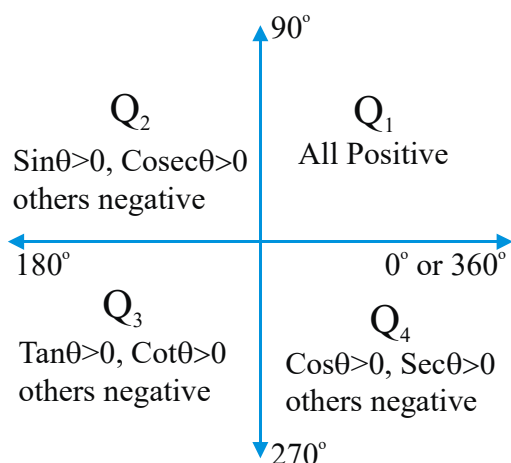
$$\begin{aligned} \sec^2 \theta - \tan^2 \theta &= 1 \Rightarrow (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1 \\ \Rightarrow \sec \theta + \tan \theta &= \frac{1}{\sec \theta - \tan \theta} \end{aligned}$$

- (ii) If $\theta \neq n\pi, n \in Z$ then

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

Trigonometric ratios of various angles:

Trig. Ratio	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0



Note : i) $\sin n\pi = \tan n\pi = \cos(2n+1)\frac{\pi}{2} = 0, \forall n \in \mathbb{Z}$

ii) $\sin(2n+1)\frac{\pi}{2} = (-1)^n$ and $\cos n\pi = (-1)^n, \forall n \in \mathbb{Z}$

➤ **Domain and range of trigonometric functions :**

Trig. function	Domain	Range
sin	$\mathbb{R}$	$[-1, 1]$
cos	$\mathbb{R}$	$[-1, 1]$
tan	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$\mathbb{R}$
cot	$\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$	$\mathbb{R}$
cosec	$\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$	$(-\infty, -1] \cup [1, \infty)$
sec	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$	$(-\infty, -1] \cup [1, \infty)$

W.E-1: If $\sin \theta = \frac{a-4}{2}$ then a lies in

Sol: $-1 \leq \sin \theta \leq 1 \Rightarrow -1 \leq \frac{a-4}{2} \leq 1$

$\Rightarrow -2 \leq a-4 \leq 2 \Rightarrow 2 \leq a \leq 6 \Rightarrow a \in [2, 6]$

➤ **Some useful results :**

(a) If $A+B=90^\circ$ or 270° , then

(i) $\sin^2 A + \sin^2 B = 1$ (ii) $\cos^2 A + \cos^2 B = 1$

(iii) $\tan A \cdot \tan B = 1$ (iv) $\cot A \cdot \cot B = 1$

(b) If $A+B=180^\circ$, then

(i) $\cos A + \cos B = 0$ (ii) $\sin A - \sin B = 0$

(iii) $\tan A + \tan B = 0$

(c) If $A+B=360^\circ$, then

(i) $\sin A + \sin B = 0$ (ii) $\cos A - \cos B = 0$

(iii) $\tan A + \tan B = 0$

W.E-2: $\tan 130^\circ \cdot \tan 140^\circ =$

Sol: $130^\circ + 140^\circ = 270^\circ$

$\therefore \tan 130^\circ \cdot \tan 140^\circ = 1$

W.E-3: $\sin^2 55^\circ + \sin^2 35^\circ =$

Sol: $55^\circ + 35^\circ = 90^\circ$

$\therefore \sin^2 55^\circ + \sin^2 35^\circ = 1$

➤ (i) If $a \cos \theta + b \sin \theta = c$ and

$a \sin \theta - b \cos \theta = k$ then $a^2 + b^2 = c^2 + k^2$

(ii) If $a \sec \theta + b \tan \theta = c$ and

$a \tan \theta + b \sec \theta = k$ then $a^2 - b^2 = c^2 - k^2$

(iii) If $a \operatorname{cosec} \theta + b \cot \theta = c$ and

$a \cot \theta + b \operatorname{cosec} \theta = k$ then $a^2 - b^2 = c^2 - k^2$

W.E-4: If $8 \cos \theta + 6 \sin \theta = 5$ then $8 \sin \theta - 6 \cos \theta =$

Sol: Let $8 \sin \theta - 6 \cos \theta = k$

$a^2 + b^2 = c^2 + k^2$

$\therefore 8^2 + 6^2 = 5^2 + k^2$

$\Rightarrow k^2 = 75 \Rightarrow k = \pm 5\sqrt{3}$

➤ $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$

for example the value of

$\sqrt{\cos^2 100^\circ} = |\cos 100^\circ|$

$= -\cos 100^\circ (\because \cos 100^\circ < 0)$

➤ (i) $\sin \theta + \sin(\pi + \theta) + \sin(2\pi + \theta) + \dots$

$\dots + \sin(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \sin \theta & \text{if } n \text{ is even} \end{cases}$

(ii) $\cos \theta + \cos(\pi + \theta) + \cos(2\pi + \theta) + \dots$

$\dots + \cos(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \cos \theta & \text{if } n \text{ is even} \end{cases}$

W.E-5: $\sin \theta + \sin(\pi + \theta) + \dots + \sin(100\pi + \theta) =$

Sol: $n = 100$ is even

$\therefore$ the required value $= \sin \theta$

W.E-6: $\cos \theta + \cos(\pi + \theta) + \dots + \cos(299\pi + \theta) =$

Sol: $n = 299$ is odd

$\therefore$ required value $= 0$

➤ If $a > 0, b > 0$ and $f(x) > 0$ then

$$af(x) + \frac{b}{f(x)} \geq 2\sqrt{ab}$$

Proof: We know that A.M. $\geq$ G.M.

$$\Rightarrow \frac{af(x) + \frac{b}{f(x)}}{2} \geq \sqrt{af(x) \cdot \frac{b}{f(x)}}$$

$$\Rightarrow af(x) + \frac{b}{f(x)} \geq 2\sqrt{ab}$$

➤ **Note:** (i) $x + \frac{1}{x} \geq 2$, for $x > 0$

(ii) $a \tan x + b \cot x \geq 2\sqrt{ab}$

(iii) $25 \cos x + 16 \sec x \geq 2\sqrt{25 \times 16} = 40$

➤ (i) $\cot^2 \theta - \cos^2 \theta = \cot^2 \theta \cos^2 \theta$

(ii) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$

(iii) $\cos^2 \theta + \sec^2 \theta = \cos^2 \theta \cdot \sec^2 \theta$

➤ (i) $\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$

(ii) $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$

(iii) $\sin^2 \theta + \cos^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$

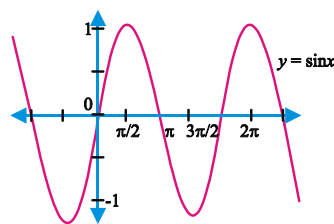
(iv) $\cos^2 \theta + \sin^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$

W.E-7: $\frac{2 + \sin^6 \theta + \cos^6 \theta}{\sin^2 \theta + \cos^4 \theta} =$

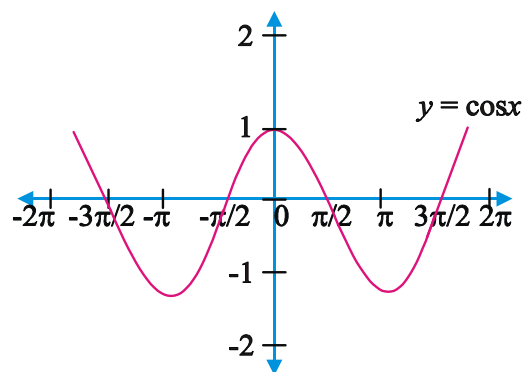
Sol:
$$\frac{2 + (1 - 3\sin^2 \theta \cos^2 \theta)}{1 - \sin^2 \theta \cos^2 \theta}$$

$$= \frac{3(1 - \sin^2 \theta \cos^2 \theta)}{(1 - \sin^2 \theta \cos^2 \theta)} = 3$$

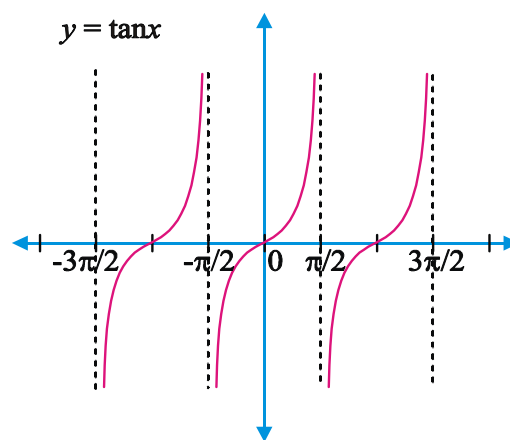
➤ **Graph of $\sin x$:**



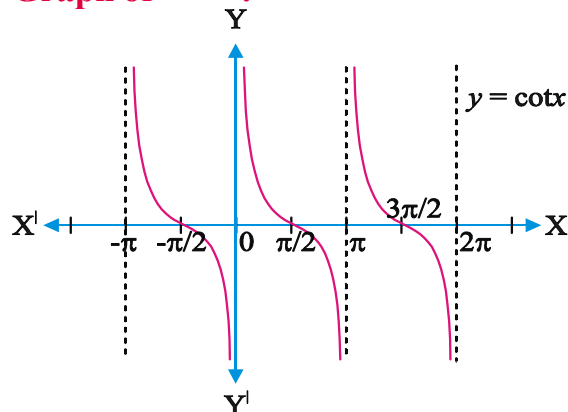
➤ **Graph of $\cos x$:**



➤ **Graph of $\tan x$:**



➤ **Graph of $\cot x$:**





LEVEL - I (C.W)

1. $-13\pi/6$ radians =
1) -390° 2) -490° 3) -410° 4) -30°
2. $\cot(1358^\circ) + \tan(3608^\circ) =$
1) -1 2) 0 3) 1 4) 2
3. $\frac{\sin(-660^\circ) \tan(1050^\circ) \sec(-420^\circ)}{\cos(225^\circ) \cos \operatorname{ec}(315^\circ) \cos(510^\circ)} =$
1) $\frac{\sqrt{3}}{4}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{2}{\sqrt{3}}$ 4) $\frac{4}{\sqrt{3}}$
4. $\sin 48^\circ \sec 42^\circ + \cos 48^\circ \cos \operatorname{ec} 42^\circ =$
1) 0 2) 2 3) 1 4) -1
5. $\log \tan 17^\circ + \log \tan 37^\circ + \log \tan 53^\circ + \log \tan 73^\circ =$
1) 0 2) 1 3) 2 4) 3
6. $\cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ =$
1) -1 2) 0 3) 1 4) 2
7. $\sec A + \tan A = 3 \Rightarrow \sec A =$
1) $\frac{10}{3}$ 2) $\frac{5}{3}$ 3) $\frac{2}{3}$ 4) $\frac{4}{3}$
8. $\sec \theta - \tan \theta = 3 \Rightarrow \theta$ lies in the quadrant
1) I 2) II 3) III 4) IV
9. $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ =$
1) 0 2) 1 3) -1 4) 2
10. $3[\sin x - \cos x]^4 + 6[\sin x + \cos x]^2 + 4[\sin^6 x + \cos^6 x] =$
1) 3 2) 6 3) 4 4) 13
11. $\frac{\sin^2 \alpha}{1 + \cot^2 \alpha} + \frac{\tan^2 \alpha}{(1 + \tan^2 \alpha)^2} + \cos^2 \alpha =$
1) -1 2) 0 3) 1 4) 2
12. $1 - \frac{\sin^2 x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x} =$
1) $\sin x$ 2) $\cos x$
3) $\operatorname{cosec} x$ 4) $\sec x$
13. If A, B, C are the angles of a triangle ABC

then $\cos\left(\frac{3A+2B+C}{2}\right) + \cos\left(\frac{A-C}{2}\right) =$

- 1) 0 2) 1 3) $\cos A$ 4) $\cos C$
14. $\frac{\cot \theta + \cos \operatorname{ec} \theta - 1}{\cot \theta - \cos \operatorname{ec} \theta + 1} =$
1) $\frac{\sin \theta}{1 + \cos \theta}$ 2) $\frac{\sin \theta}{1 - \cos \theta}$
3) $\frac{1 - \cos \theta}{\sin \theta}$ 4) $\frac{\cos \theta}{1 - \sin \theta}$
 15. $\sin^2(51^\circ - x) + \sin^2(39^\circ + x) =$
1) -1 2) 0 3) 1 4) 2
 16. $\frac{\sin^2 \frac{\pi}{18} + \sin^2 \frac{\pi}{9} + \sin^2 \frac{7\pi}{18} + \sin^2 \frac{4\pi}{9}}{\cos^2 \frac{\pi}{18} + \cos^2 \frac{\pi}{9} + \cos^2 \frac{7\pi}{18} + \cos^2 \frac{4\pi}{9}} =$
1) 1 2) 2 3) 3 4) 4
 17. If ABCD is a cyclic quadrilateral, then
 $\cos(180^\circ + A) + \cos(180^\circ - B) + \cos(180^\circ - C) - \sin(90^\circ - D) =$
1) -1 2) 0 3) 1 4) 2
 18. $\left[\sin\left(\frac{\pi}{2} - x\right) + \sin(\pi - x) \right]^2 + \left[\cos\left(\frac{3\pi}{2} - x\right) + \cos(2\pi - x) \right]^2 =$
1) 0 2) 2 3) 4 4) 8
 19. $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3 \Rightarrow \cos \theta_1 + \cos \theta_2 + \cos \theta_3 =$
1) 0 2) 1 3) 2 4) 3
 20. $8 \sin^2 x + 3 \cos^2 x = 5 \Rightarrow \cot x =$
1) $\pm \frac{1}{\sqrt{2}}$ 2) $\pm \frac{1}{\sqrt{3}}$ 3) $\pm \sqrt{\frac{3}{2}}$ 4) $\pm \sqrt{\frac{2}{3}}$
 21. $\sin x + \sin^2 x = 1 \Rightarrow \cos^2 x + \cos^4 x =$
1) 0 2) 1 3) 2 4) -1
 22. If $\cos(x - y) = -1$, then $\cos x + \cos y =$
1) 0 2) 1 3) 2 4) -1

23. If $\sin x + \operatorname{cosec} x = 2$ then $\sin^8 x + \operatorname{cosec}^8 x =$
 1) 1 2) 2 3) 3 4) 4
24. If $\sin x + \cos x = a$ then $|\sin x - \cos x| =$
 1) $\sqrt{1-a^2}$ 2) $\sqrt{a^2-1}$ 3) $\sqrt{2-a^2}$ 4) $\sqrt{a^2-2}$
25. $(1 + \sin A)(1 + \sin B)(1 + \sin C) = (1 - \sin A)$
 $(1 - \sin B)(1 - \sin C) = k \Rightarrow k =$
 1) $+\sin A \sin B \sin C$
 2) $\pm \cos A \cos B \cos C$
 3) $\pm \sec A \sec B \sec C$
 4) $\pm \operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$
26. $\operatorname{cosec}^2 A \cot^2 A - \sec^2 A \tan^2 A -$
 $(\cot^2 A - \tan^2 A)(\sec^2 A \operatorname{cosec}^2 A - 1) =$
 1) 0 2) 1 3) -1 4) 2
27. $\tan^2 \theta + \sec \theta = 5 \Rightarrow \sec \theta =$
 1) 3 2) 2 3) 1 4) -1
28. $\log \sin 1^\circ \cdot \log \sin 2^\circ \dots \dots \log \sin 179^\circ$
 1) 1 2) 0 3) -1 4) 2
29. If $\tan(\alpha + \beta) = \sqrt{3}$, $\tan(\alpha - \beta) = 1$ then
 $\tan 6\beta =$
 1) -1 2) 0 3) 1 4) 2
30. If $x = r \cos \theta \cos \phi$, $y = r \cos \theta \sin \phi$,
 $z = r \sin \theta$ then $x^2 + y^2 + z^2 =$
 1) r^2 2) y^2 3) x^2 4) z^2
31. $\sin^2 A \cos^2 B + \cos^2 A \sin^2 B$
 $+ \sin^2 A \sin^2 B + \cos^2 A \cos^2 B =$
 1) -1 2) 0 3) 1 4) 2
32. If $2 \sin^2 A - \cot^2 B = 1$ then $\sin A \sin B =$
 1) $-\frac{1}{\sqrt{2}}$ 2) $\pm \frac{1}{\sqrt{2}}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{-1}{2\sqrt{2}}$
33. The value of
 $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$ is
 1) 0 2) 1 3) 2 4) 3
34. If $x \tan^2 120^\circ + 4 \cos^2 150^\circ = 9$ then $x =$
 1) 3 2) 1 3) 2 4) 4
35. In $\triangle ABC$, right angled at C, then
 $\tan A + \tan B =$
 1) $\frac{a^2}{bc}$ 2) $\frac{b^2}{ac}$ 3) $\frac{c^2}{ab}$ 4) $\frac{ab}{c}$
36. If $a = \sin 170^\circ + \cos 170^\circ$ then
 1) $a > 0$ 2) $a < 0$ 3) $a = 0$ 4) $a = 1$
37. $\tan \frac{\pi}{24} \tan \frac{3\pi}{24} \tan \frac{5\pi}{24} \tan \frac{7\pi}{24} \tan \frac{9\pi}{24} \tan \frac{11\pi}{24} \tan \frac{13\pi}{24} \tan \frac{15\pi}{24} =$
 1) 1 2) -1 3) $\sqrt{3}$ 4) $-\sqrt{3}$
38. $x = a \sec^3 \theta \tan \theta$, $y = b \tan^3 \theta \sec \theta$
 then $\sin^2 \theta =$
 1) $\frac{x}{a} - \frac{y}{b}$ 2) $\frac{x}{a} + \frac{y}{b}$ 3) $\frac{xy}{ab}$ 4) $\frac{ay}{bx}$
39. If $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$, $\frac{x}{a} \sin \theta - \frac{y}{b} \cos \theta = 1$
 then $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$
 1) 1 2) -1 3) 2 4) 3
40. In a $\triangle ABC$, if $\cot A \cot B \cot C > 0$ then the
 triangle is
 1) acute angled 2) right angled
 3) obtuse angled 4) can't be decided
41. If α, β are complementary angles and
 $\sin \alpha = \frac{3}{5}$, then $\sin \alpha \cos \beta - \cos \alpha \sin \beta =$
 1) $\frac{7}{25}$ 2) $-\frac{7}{25}$ 3) $\frac{25}{7}$ 4) $-\frac{25}{7}$
42. $4a^2 \sin^2 \left(\frac{3\pi}{4} \right) - 3[a \tan 225^\circ]^2 + [2a \cos 315^\circ]^2 =$
 1) 0 2) a 3) $\sqrt{2}a$ 4) a^2
43. If θ lies in the first quadrant and $5 \tan \theta = 4$, then
 $\frac{5 \sin \theta - 3 \cos \theta}{\sin \theta + 2 \cos \theta} =$ (EAM-2007)
 1) 5/14 2) 3/14 3) 1/14 4) 0
44. If $a \sin^2 \theta + b \cos^2 \theta = c$, then $\tan^2 \theta =$
 (EAM-2010)
 1) $\frac{b-c}{a-c}$ 2) $\frac{c-b}{a-c}$ 3) $\frac{a-c}{b-c}$ 4) $\frac{a-c}{c-b}$

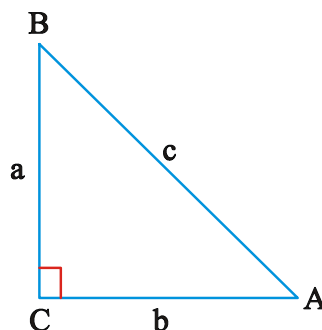
LEVEL-I(C.W)-KEY

- 01) 1 02) 2 03) 3 04) 2 05) 1
 06) 2 07) 2 08) 4 09) 1 10) 4
 11) 3 12) 2 13) 1 14) 2 15) 3
 16) 1 17) 2 18) 2 19) 1 20) 3
 21) 2 22) 1 23) 2 24) 3 25) 2
 26) 1 27) 2 28) 2 29) 3 30) 1
 31) 3 32) 2 33) 2 34) 3 35) 3
 36) 2 37) 4 38) 4 39) 3 40) 1
 41) 2 42) 4 43) 1 44) 2

LEVEL-I(C.W)-HINTS

- $-13 \times \frac{\pi}{6} \times \frac{180}{\pi} = -390^\circ$
- $1358^\circ = 8(180^\circ) - 82^\circ$,
 $3608^\circ = 20(180^\circ) + 8^\circ$
- Convert the corresponding values.
- $\sin 48^\circ \operatorname{cosec} 48^\circ + \cos 48^\circ \sec 48^\circ = 1 + 1 = 2$
- $\therefore \text{If } A + B = 90^\circ \Rightarrow \tan A \tan B = 1$
- $\cos(204^\circ) = \cos(180^\circ + 24^\circ)$
 $\cos(125^\circ) = \cos(180^\circ - 55^\circ)$
- Use $(\sec^2 A - \tan^2 A) = 1$
- Use $(\sec^2 A - \tan^2 A) = 1$
- $A + B = 360^\circ \Rightarrow \sin A + \sin B = 0$
- Put $x = 90^\circ$
- Put $\alpha = 45^\circ$
- Let $x = 90^\circ$
- $A = B = C = 60^\circ$
- $1 = \operatorname{cosec}^2 \theta - \cot^2 \theta$
- $A + B = 90^\circ \Rightarrow \sin^2 A + \sin^2 B = 1$
- If $A + B = 90^\circ$ then $(\cos^2 A + \cos^2 B) = 1$
 and $(\sin^2 A + \sin^2 B) = 1$
- $A + C = 180^\circ$, $B + D = 180^\circ$
- Simplify
- Put $\theta_1 = \theta_2 = \theta_3 = 90^\circ$
- Divide the given equation by $\sin^2 x$ both sides
- $\sin x = (1 - \sin^2 x) \Rightarrow \sin x = \cos^2 x$

- $x - y = \pi$
- Put $x = 90^\circ$
- $\alpha = \frac{\pi}{4}$
- $k^2 = (1 + \sin A)(1 + \sin B)(1 + \sin C)(1 - \sin A)$
 $(1 - \sin B)(1 - \sin C)$
 $K^2 = (1 - \sin^2 A)(1 - \sin^2 B)(1 - \sin^2 C)$
 $K^2 = \cos^2 A \cos^2 B \cos^2 C$
 $K = \pm \cos A \cos B \cos C$
- Put $A = 45^\circ$
- $\tan^2 \theta + \sec \theta = 5, \Rightarrow (\sec^2 \theta - 1) + \sec \theta = 5,$
 $\Rightarrow \sec^2 \theta + \sec \theta - 6 = 0, \sec \theta = 2 \text{ or } -3$
- $\log(\sin 90^\circ) = 0$
- Let $(\alpha + \beta) = 60^\circ$
 $(\alpha - \beta) = 45^\circ \Rightarrow 2\beta = 15^\circ \Rightarrow 6\beta = 45^\circ$
- Square and add
- $\sin^2 A(\cos^2 B + \sin^2 B) + \cos^2 A(\sin^2 B + \cos^2 B)$ or
 Put $A = B = 0^\circ$
- Put $\cot B = \frac{\cos B}{\sin B}$
- use $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$
- Substitute the corresponding values.
-



$$\angle C = 90^\circ$$

$$\therefore \tan A + \tan B = \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$$

36. $a = \sin 170^\circ + \cos 170^\circ = \sin 10^\circ - \cos 10^\circ < 0$

37. $A + B = 90^\circ \Rightarrow \tan A \tan B = 1$

38. $\frac{y}{x} = \frac{b \tan^3 \theta \sec \theta}{a \sec^3 \theta \tan \theta} = \frac{b}{a} \sin^2 \theta$

39. by squaring and adding $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

40. $\cot A \cot B \cot C > 0$
 $\Rightarrow \cot A, \cot B, \cot C$ are positive
 $\Rightarrow \Delta ABC$ is acute angled

41. $\sin \alpha = \frac{3}{5}, \cos \beta = \frac{3}{5}, \cos \alpha = \frac{4}{5}, \sin \beta = \frac{4}{5}$

42. Given $= 4a^2 \left(\frac{1}{\sqrt{2}} \right)^2 - 3[a \times 1]^2 + \left[2a \times \frac{1}{\sqrt{2}} \right]^2$

43. Divide the Nr. and Dr. with $\cos \theta$

44. Divide throughout with $\cos^2 \theta$
 $\Rightarrow a \tan^2 \theta + b = c \sec^2 \theta \Rightarrow (a-c) \tan^2 \theta = (c-b)$



LEVEL - I (H.W)

1. $-11\pi/3$ radians =

1) -390° 2) -620° 3) -610° 4) -660°

2. $\sin 4530^\circ =$

1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) $\frac{\sqrt{3}}{2}$ 4) $-\frac{\sqrt{3}}{2}$

3. $\frac{\sin 150^\circ - 5 \cos 300^\circ + 7 \tan 225^\circ}{\tan 135^\circ + 3 \sin 210^\circ} =$

1) 10 2) -5 3) -2 4) -3/2

4. $\cos 23^\circ \operatorname{cosec} 67^\circ - \sin 23^\circ \sec 67^\circ =$

1) 0 2) 2 3) 1 4) -1

5. $\log \tan 18^\circ + \log \tan 36^\circ + \log \tan 54^\circ + \log \tan 72^\circ =$

1) $\log 4$ 2) $\log 3$ 3) $\log 2$ 4) 0

6. $\cos 5^\circ + \cos 24^\circ + \cos 175^\circ + \cos 204^\circ + \cos 300^\circ =$

1) $1/2$ 2) 1 3) $3/2$ 4) 2

7. $\operatorname{cosec} A + \cot A = \frac{2}{3} \Rightarrow \cos A =$

1) $\frac{5}{13}$ 2) $\frac{13}{5}$ 3) $-\frac{5}{13}$ 4) $-\frac{13}{5}$

8. If $\operatorname{cosec} \theta - \cot \theta = 5$ then θ lies in the quadrant

1) I 2) II 3) III 4) IV

9. $\tan 20^\circ + \tan 40^\circ + \tan 60^\circ + \dots + \tan 180^\circ =$

1) 0 2) 1 3) 2 4) 3

10. $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\sec \alpha + \cos \alpha)^2 =$

$k + \tan^2 \alpha + \cot^2 \alpha \Rightarrow k =$

1) 9 2) 7 3) 5 4) 3

11. $\frac{1}{(1 + \cot^2 \alpha)^2} + \frac{\tan^2 \alpha}{(1 + \tan^2 \alpha)^2} + \frac{1}{1 + \tan^2 \alpha} =$

1) -1 2) 0 3) 1 4) 2

12. $\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} + \frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} = k$ then $k =$

1) 0 2) 1 3) 2 4) -1

13. In ΔABC , $\tan(A - B - C) =$

1) $\sin 2A$ 2) 1 3) $\tan 2A$ 4) 0

14. $\frac{1 + \cot \alpha + \operatorname{cosec} \alpha}{1 - \cot \alpha + \operatorname{cosec} \alpha} =$

1) $\frac{\sin \alpha}{1 + \cos \alpha}$ 2) $\frac{\sin \alpha}{1 - \cos \alpha}$

3) $\frac{1 + \cos \alpha}{\sin \alpha}$ 4) $\frac{1 - \sin \alpha}{\cos \alpha}$

15. $\cos^2(125^\circ - x) - \cos^2(55^\circ + x) =$

1) -1 2) 0 3) 1 4) 2

16. $\sin^2 \frac{\pi}{18} + \sin^2 \frac{2\pi}{18} + \sin^2 \frac{4\pi}{18} + \sin^2 \frac{8\pi}{18}$

$+ \sin^2 \frac{7\pi}{18} + \sin^2 \frac{5\pi}{18} =$

1) 1 2) 2 3) 3 4) 4

17. If ABCD is a quadrilateral then

$\tan\left(\frac{A+B}{4}\right) =$

1) $\cos\left(\frac{C-D}{4}\right)$ 2) $\cot\left(\frac{C-D}{4}\right)$

3) $\cos\left(\frac{C+D}{4}\right)$ 4) $\cot\left(\frac{C+D}{4}\right)$

18. $\left[\cos\left(\frac{\pi}{2}-x\right) + \cos(\pi-x) \right] + \left[\sin\left(\frac{3\pi}{2}-x\right) + \sin(2\pi-x) \right] =$

- 1) 0 2) 2 3) 1 4) 4

19. If $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 + \cos \theta_4 + \cos \theta_5 = 5$

then $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4 + \sin \theta_5 =$

- 1) 3 2) 2 3) 1 4) 0

20. $9 \cos^2 x + 4 \sin^2 x = 5 \Rightarrow \tan x =$

- 1) ± 1 2) ± 2 3) ± 3 4) ± 4

21. $\cos x + \cos^2 x = 1 \Rightarrow \sin^8 x + 2 \sin^6 x + \sin^4 x =$

- 1) 0 2) 1 3) 2 4) -1

22. If $\cot(\alpha + \beta) = 0$ then $\sin(\alpha + 2\beta) =$

- 1) $\sin \alpha$ 2) $\cos \alpha$ 3) $\sin \beta$ 4) $\cos \beta$

23. If $\tan \alpha + \cot \alpha = 2$ then $\sqrt{\tan \alpha} + \sqrt{\cot \alpha} =$

- 1) $\sqrt{2}$ 2) $2\sqrt{2}$ 3) 2 4) $4\sqrt{2}$

24. $5 \sin x + 4 \cos x = 3 \Rightarrow 4 \sin x - 5 \cos x =$

- 1) 4 2) $4\sqrt{2}$ 3) $3\sqrt{2}$ 4) $\sqrt{2}$

25. If $k = (\sec A + \tan A)(\sec B + \tan B)(\sec C + \tan C) =$

$(\sec A - \tan A)(\sec B - \tan B)(\sec C - \tan C)$

then $k =$

- 1) 0 2) ± 1 3) ± 3 4) ± 4

26. The value of $\cot^2 \alpha \left(\frac{\sec \alpha - 1}{1 + \sin \alpha} \right)$

$+ \sec^2 \alpha \left(\frac{\sin \alpha - 1}{1 + \sec \alpha} \right)$ is

- 1) 0 2) 1 3) 2 4) -2

27. $\sec^2 \theta + \tan \theta = 13 \Rightarrow \tan \theta =$

- 1) 4 2) 3 3) -3 4) 5

28. $\log(\tan 1^\circ) \log(\tan 2^\circ) \log(\tan 3^\circ) \dots \log(\tan 45^\circ) =$

- 1) 0 2) 1 3) -1 4) $\frac{1}{2}$

29. If $\sin(\alpha + \beta) = 1$, $\sin(\alpha - \beta) = \frac{1}{2}$ then

$\tan(\alpha + 2\beta) \tan(2\alpha + \beta) =$

- 1) 1 2) -1 3) 0 4) 2

30. If $x = r \cos \alpha \cos \beta \cos \gamma$;

$y = r \cos \alpha \cos \beta \sin \gamma$;

$z = r \sin \alpha \cos \beta$; $\mu = r \sin \beta$ then

$x^2 + y^2 + z^2 + \mu^2 =$

- 1) r 2) $2r$ 3) r^2 4) $4r^2$

31. $\sec^2 A \sec^2 B - \sec^2 A \tan^2 B - \tan^2 A \sec^2 B + \tan^2 A \tan^2 B =$

- 1) 1 2) 0 3) -1 4) 2

32. $2 \cos^2 B - 1 = \tan^2 A$, then $\cos A \cos B =$

- 1) $\pm \frac{1}{2}$ 2) $\pm \frac{1}{3}$ 3) $\pm \frac{1}{\sqrt{2}}$ 4) ± 1

33. $\sin^4 \theta + 2 \sin^2 \theta \left(1 - \frac{1}{\cos^2 \theta} \right) + \cos^4 \theta =$

- 1) 1 2) 0 3) $\frac{1}{2}$ 4) -1

34. If $x \cot^2 120^\circ + 4 \sin^2 150^\circ = 3$, then $x =$

- 1) 2 2) 7 3) 6 4) 5

35. In a right angled triangle ABC, $\angle C = 90^\circ$, then

$\cos^2 A + \cos^2 B =$

- 1) 2 2) 1 3) $\frac{1}{2}$ 4) $\frac{3}{4}$

36. If $x = \sin 130^\circ + \cos 130^\circ$ then

- 1) $x < 0$ 2) $x = 0$ 3) $x > 0$ 4) $x \geq 0$

37. $\cot \frac{\pi}{20} \cdot \cot \frac{3\pi}{20} \cdot \cot \frac{5\pi}{20} \cdot \cot \frac{7\pi}{20} \cdot \cot \frac{9\pi}{20} \cdot \cot \frac{15\pi}{20} =$

- 1) 1 2) -1 3) $\sqrt{3}$ 4) $-\sqrt{3}$

38. If $x = a \sec^n \theta$; $y = b \tan^n \theta$ then

$\left(\frac{x}{a} \right)^{\frac{2}{n}} - \left(\frac{y}{b} \right)^{\frac{2}{n}} =$

- 1) 0 2) -1 3) 1 4) 2

39. If $x = h + p \sec \alpha$, $y = k + q \sec \alpha$ then

$\left(\frac{p}{x-h} \right)^2 + \left(\frac{q}{y-k} \right)^2 =$

- 1) 1 2) -1 3) 0 4) $\frac{1}{2}$

40. If A, B, C are angles of a triangle such that A is obtuse then

- 1) $\tan A \tan B > 1$ 2) $\tan B \tan C < 1$
3) $\tan C \tan A > 1$ 4) $\tan A \tan B \tan C > 1$

41. If α, β are complementary angles,

$$\sin \alpha = \frac{3}{5}, \text{ then } \cos \alpha \cos \beta - \sin \alpha \sin \beta =$$

- 1) 1 2) 2 3) $\frac{4}{5}$ 4) 0

42. $a^2 \cos^2 \frac{2\pi}{3} - 4a^2 \tan^2 \frac{3\pi}{4} + 2a^2 \sin^2 \frac{2\pi}{3} =$

- 1) a^2 2) 0 3) $\frac{3a^2}{4}$ 4) $\frac{-9a^2}{4}$

43. If $\tan \theta = \frac{p}{q}$ then $\frac{p \sin \theta - q \cos \theta}{p \sin \theta + q \cos \theta} =$

- 1) $\frac{2p}{p^2 + q^2}$ 2) $\frac{2pq}{p^2 + q^2}$

- 3) $\frac{p^2 - q^2}{p^2 + q^2}$ 4) $\frac{q^2 - p^2}{p^2 + q^2}$

44. If $a \sec \theta + b \tan \theta = c$ then

$$(a \tan \theta + b \sec \theta)^2 =$$

- 1) $a^2 + b^2 + c^2$ 2) $-a^2 + b^2 + c^2$
3) $a^2 - b^2 + c^2$ 4) $a^2 + b^2 - c^2$.

LEVEL-I-(H.W)-KEY

- 01) 4 02) 2 03) 3 04) 1 05) 4
06) 1 07) 3 08) 2 09) 1 10) 2
11) 3 12) 3 13) 3 14) 3 15) 2
16) 3 17) 4 18) 2 19) 4 20) 2
21) 2 22) 4 23) 3 24) 2 25) 2
26) 1 27) 2 28) 1 29) 1 30) 3
31) 1 32) 3 33) 1 34) 3 35) 2
36) 3 37) 2 38) 3 39) 1 40) 2
41) 4 42) 4 43) 3 44) 2

LEVEL-I-(H.W)-HINTS

- $-11 \frac{\pi}{3} \times \frac{180}{\pi} = -660^\circ$
- $\sin(4530^\circ) = \sin((12)360^\circ + 210^\circ)$
 $= \sin 210^\circ = -\sin 30^\circ = \frac{1}{2}$
- Convert the corresponding values.

4. $\cos 23^\circ \operatorname{cosec}(90^\circ - 23^\circ) - \sin 23^\circ \sec(90^\circ - 23^\circ)$
 $= 1 - 1 = 0$

5. $\log(\tan 18^\circ \tan 36^\circ \tan 54^\circ \tan 72^\circ) = \log 1 = 0$
[$\because$ If $A + B = 90^\circ \Rightarrow \tan A \tan B = 1$]

6. $\alpha + \beta = 180^\circ$
 $\Rightarrow \cos \alpha + \cos \beta = 0$

7. Use $(\cos^2 \theta - \cot^2 \theta) = 1$

8. Use $(\cos^2 A - \cot^2 A) = 1$. Find the quadrant in which A lies

9. $A + B = 180^\circ \Rightarrow \tan A + \tan B = 0$

10. Square and simplify

11. Put $\alpha = 45^\circ$

12. Put $A = 0^\circ$

13. $B + C = 180^\circ - A \Rightarrow \tan(A + A - 180^\circ) = \tan 2A$

14. Use $1 = (\operatorname{cosec}^2 \alpha - \cot^2 \alpha)$

15. $(125^\circ - x) + (55^\circ + x) = 180^\circ$

16. If $(A + B) = \frac{\pi}{2}$ then $\sin^2 A + \sin^2 B = 1$

17. $(A + B + C + D) = 360^\circ \Rightarrow \frac{A+B}{4} = \frac{\pi}{2} - \frac{C+D}{4}$

18. $(\sin x - \cos x)^2 + (-\cos x - \sin x)^2 = 2$

19. Put $\theta_1 = \theta_2 = \theta_3 = 0^\circ$

20. Divide the given equation by $\cos^2 x$ both sides.

21. $\cos x = 1 - \cos^2 x \Rightarrow \cos x = \sin^2 x$

22. $\cot(\alpha + \beta) = 0 \Rightarrow \alpha + \beta = \frac{\pi}{2}$
 $\Rightarrow \sin(\alpha + 2\beta) = \sin(90^\circ + \beta) = \cos \beta$

23. Put $\alpha = \frac{\pi}{4}$

24. $a \sin x + b \cos x = c \Rightarrow b \sin x - a \cos x = \pm \sqrt{a^2 + b^2 - c^2}$

25. $K^2 = (1 + \sin A)(1 + \sin B)(1 + \sin C)(1 - \sin A)(1 - \sin B)(1 - \sin C)$
 $K^2 = (1 - \sin^2 A)(1 - \sin^2 B)(1 - \sin^2 C)$
 $K^2 = \cos^2 A \cdot \cos^2 B \cdot \cos^2 C$
 $K = \pm \cos A \cos B \cos C$

26. Put $\alpha = 45^\circ$; verify the options.
27. $1 + \tan^2 \theta + \tan \theta = 13 \Rightarrow \tan \theta = 3 \text{ or } -4$
28. $\log(\tan 45^\circ) = \log 1 = 0$
29. $\alpha + \beta = 90^\circ, \alpha - \beta = 30^\circ \Rightarrow \alpha = 60^\circ, \beta = 30^\circ$
30. Squaring and adding
31. Put $A = B = 45^\circ$
32. $2 \cos^2 B = 1 + \tan^2 A$
33. $\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$
34. $x \cdot \frac{1}{3} + 4 \cdot \frac{1}{4} = 3 \Rightarrow x = 6$
35. $A + B = 90^\circ \Rightarrow \cos^2 A + \cos^2 B = 1$
36. $x = \sin 50^\circ - \cos 50^\circ$
37. $A + B = 90^\circ \Rightarrow \cot A \cot B = 1$
38. $\sec^2 \theta - \tan^2 \theta = 1$
39. $\sin^2 \theta + \cos^2 \theta = 1$
40. A is obtuse $\Rightarrow B + C < 90^\circ \Rightarrow \tan B \tan C < 1$
41. $\sin \alpha = \frac{3}{5}, \cos \beta = \frac{3}{5}, \cos \alpha = \frac{4}{5}, \sin \beta = \frac{4}{5}$

$$42. a^2 \left(-\frac{1}{2} \right)^2 - 4a^2 (-1)^2 + 2a^2 \left(\frac{\sqrt{3}}{2} \right)^2$$

43. Divide the Nr and Dr. with $\cos \theta$

44. Let $a \tan \theta + b \sec \theta = k$

$$\therefore a^2 - b^2 = c^2 - k^2 \Rightarrow k^2 = c^2 - a^2 + b^2$$



LEVEL - II (C.W)

1. If θ is not in 4th quadrant and $\tan \theta = -4/3$ then
 $5 \sin \theta + 10 \cos \theta + 9 \sec \theta + 16 \csc \theta + 4 \cot \theta =$
 1) -1 2) 2/5 3) 4/5 4) 0
2. $f(x) = x^3 - 3x + 5$ then $f\left(\sin \frac{3\pi}{2}\right) + f\left(\cos \frac{3\pi}{2}\right) =$
 1) 10 2) 12 3) 14 4) 16
3. If $\tan \theta, 2 \tan \theta + 2, 3 \tan \theta + 3$ are in G.P then
 the value of $\frac{7 - 5 \cot \theta}{9 - 4 \sqrt{\sec^2 \theta - 1}}$ is
 1) $\frac{12}{5}$ 2) $\frac{-33}{28}$ 3) $\frac{33}{100}$ 4) $\frac{12}{13}$

4. In a triangle ABC, $C = 90^\circ$, then the equation whose roots are $\tan A, \tan B$ is

- 1) $abx^2 + c^2x + ab = 0$
- 2) $abx^2 + c^2x - ab = 0$
- 3) $abx^2 - c^2x - ab = 0$
- 4) $abx^2 - c^2x + ab = 0$

5. Which of the following is correct?

- 1) $\sin 1^\circ > \sin 1$ 2) $\sin 1^\circ < \sin 1$
- 3) $\sin 1^\circ = \sin 1$ 4) $\sin 1^\circ = \frac{\pi}{180} \sin 1$

6. If $a = \cos 3$ and $b = \sin 8$ then

- 1) $a > 0, b > 0$ 2) $ab < 0$ 3) $a > b$ 4) $ab > 0$

7. $\operatorname{cosec} A = 4p + \frac{1}{16p} \Rightarrow \operatorname{cosec} A + \cot A =$

- 1) $8p$ 2) $\frac{1}{8p}$ 3) $-8p$ (or) $\frac{1}{8p}$ 4) $8p$ (or) $\frac{1}{8p}$

8. If $\sqrt{\frac{1 - \sin A}{1 + \sin A}} = \sec A - \tan A$ then A lies in the quadrants

- 1) I, II 2) II, III 3) I, IV 4) I, III

9. If $\sin \theta + \cos \theta = m$ and $\sec \theta + \csc \theta = n$, then $n(m+1)(m-1) =$

- 1) m 2) n 3) $2m$ 4) $2n$

10. If $1 + \sin x + \sin^2 x + \dots$ to $\infty = 4 + 2\sqrt{3}, 0 < x < \pi$

and $x \neq \frac{\pi}{2}$ then $x =$

- 1) $30^\circ, 60^\circ$ 2) $60^\circ, 120^\circ$
- 3) $90^\circ, 120^\circ$ 4) $30^\circ, 45^\circ$

11. $\cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 360^\circ =$

- 1) 18 2) 27 3) 36 4) 45

12. If $x = a \cos^2 \theta \sin \theta$ and $y = a \sin^2 \theta \cos \theta$,

then $\frac{(x^2 + y^2)^3}{x^2 y^2} =$

- 1) a 2) a^3 3) a^2 4) a^5

13. $a \sec \theta + b \tan \theta = 1, a \sec \theta - b \tan \theta = 5$

$$\Rightarrow a^2(b^2 + 4) =$$

1) $3b^2$ 2) $9b^2$ 3) b^2 4) $4b^2$

14. $\tan A = a \tan B, \sin A = b \sin B \Rightarrow \frac{b^2 - 1}{a^2 - 1} =$

1) $\sin^2 A$ 2) $\sin^3 A$ 3) $\cos^2 A$ 4) $\cos^3 A$

15. If $\frac{\cos^2 \theta}{a} = \frac{\sin^2 \theta}{b}$ then $\frac{\cos^4 \theta}{a} + \frac{\sin^4 \theta}{b} =$

1) $\frac{1}{a+b}$ 2) $\frac{1}{(a+b)^2}$

3) $\frac{1}{a^2} + \frac{1}{b^2}$ 4) $a+b$

16. If $a = x \cos^2 A + y \sin^2 A$ then $(x-a)(y-a)$

$$+ (x-y)^2 \sin^2 A \cos^2 A =$$

1) 0 2) 1 3) $xy + a^2$ 4) $xy - a^2$

17. If $a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha = m$ and

$$a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha = n$$
 then

$$(m+n)^{2/3} + (m-n)^{2/3} =$$

1) $2a^2$ 2) $2a^{1/3}$ 3) $2a^{2/3}$ 4) $2a^3$

18. $\cos^4 \alpha - \sin^4 \alpha = a$ then $\frac{1-a}{1+a} =$

1) $\tan^2 \alpha$ 2) $\cot^2 \alpha$ 3) $-\tan^2 \alpha$ 4) $-\cot^2 \alpha$

19. $a = \frac{1 + \sin x}{1 - \cos x + \sin x} \Rightarrow \frac{1 + \cos x + \sin x}{2 \sin x} =$

1) a 2) $\frac{1}{a}$ 3) a^2 4) $\frac{1}{a^2}$

20. Eliminate θ from $x = 1 + \tan \theta, y = 2 + \cot \theta$

1) $xy + 1 = x + y$ 2) $xy + 2 = 2x + y$

3) $xy + 1 = 2x + y$ 4) $xy + 1 = 2y + x$

21. $f(x) = \sin^2 x + \operatorname{cosec}^2 x \Rightarrow$

1) $f(x) < 1$ 2) $f(x) = 1$

3) $1 < f(x) < 2$ 4) $f(x) \geq 2$

22. Which of the following is not possible ?

1) $\sin \theta = \frac{5}{7}$ 2) $\cos \theta = \frac{1+a^2}{1-a^2}, |a| \neq 1$

3) $\tan \theta = 100$ 4) $\sec \theta = \frac{5}{2}$

23. If $\theta = \frac{11\pi}{6}$, then $\cos \theta + \sin \theta =$

1) $\frac{\sqrt{3}+1}{\sqrt{2}}$ 2) $\frac{\sqrt{3}-1}{\sqrt{2}}$ 3) $\frac{\sqrt{3}-1}{2}$ 4) $\frac{\sqrt{3}+1}{2}$

24. The value of $\sin\left(n\pi + (-1)^n \frac{\pi}{4}\right), n \in I$, is

1) 0 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{-1}{\sqrt{2}}$ 4) $\frac{\sqrt{3}}{2}$

25. If $e^{(\sin^2 x + \sin^4 x + \sin^6 x + \dots \infty) \log 2} = 8$ and

$$0 < x < \frac{\pi}{2} \text{ then } \frac{\cos x}{\cos x + \sin x} =$$

1) $\frac{\sqrt{3}+1}{2}$ 2) $\frac{\sqrt{3}-1}{2}$ 3) $\frac{2}{\sqrt{3}+1}$ 4) $\frac{2}{\sqrt{3}-1}$

26. If $\cos x = \tan y, \cot y = \tan z$ and $\cot z = \tan x$, then $\sin x =$ (EAM-20014)

1) $\frac{\sqrt{5}+1}{4}$ 2) $\frac{\sqrt{5}-1}{4}$ 3) $\frac{\sqrt{5}+1}{2}$ 4) $\frac{\sqrt{5}-1}{2}$

27. If $\sin \theta + \cos \theta = p$ and $\sin^3 \theta + \cos^3 \theta = q$, then $p(p^2 - 3) =$ (EAM-2013)

1) q 2) $2q$ 3) $-q$ 4) $-2q$

28. If $0 \leq x \leq \pi, 81^{\sin^2 x} + 81^{\cos^2 x} = 30$ then $x =$

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{15}$ 4) $\frac{\pi}{8}$

29. $\operatorname{cosec}^2 \alpha = \frac{4xy}{(x+y)^2} [(x, y) \neq (0, 0)]$ is

1) Possible for $x = y$ 2) Impossible for $x = y$
3) Possible for $x = -y$ 4) Not possible

30. If $a \sin \theta + b \cos \theta = c$ then $\frac{a - b \tan \theta}{b + a \tan \theta} =$

$$1) \frac{\pm\sqrt{b^2 - c^2 + a^2}}{c} \quad 2) \frac{\pm\sqrt{b^2 + c^2 - a^2}}{c}$$

$$3) \frac{\pm\sqrt{b^2 - c^2 - a^2}}{c} \quad 4) \frac{\pm\sqrt{a^2 + b^2 + c^2}}{b^2 - c^2}$$

LEVEL-II(C.W)-KEY

- 01) 4 02) 2 03) 2 04) 4 05) 2
 6) 2 7) 4 8) 3 9) 3 10) 2
 11) 3 12) 3 13) 2 14) 3 15) 1
 16) 1 17) 3 18) 1 19) 1 20) 3
 21) 4 22) 2 23) 3 24) 2 25) 2
 26) 4 27) 4 28) 1 29) 1 30) 1

LEVEL-II(C.W)-HINTS

- θ lies in II quadrant, substitute the values
- $\sin 270^\circ = -1$ & $\cos 270^\circ = 0$
- $(2 \tan \theta + 2)^2 = \tan \theta (3 \tan \theta + 3)$
 $\Rightarrow \tan^2 \theta + 5 \tan \theta + 4 = 0$
 $\Rightarrow \tan \theta = -1, -4$
 but $\tan \theta = -4$ only satisfies above condition
- $\tan A = \frac{a}{b}$, $\tan B = \frac{b}{a}$ and $(a^2 + b^2) = c^2$
 Find sum and product of roots
- $1 > 1^0$
- 3° lies in the second quadrant
 $\therefore \cos 3 < 0$
 8° lies in the second quadrant
 $\therefore \sin 8 > 0 \quad \Rightarrow ab < 0$
- Use $(\operatorname{cosec}^2 A - \cot^2 A) = 1$
 If $\operatorname{cosec} A = \left(4P + \frac{1}{16P}\right)$
 then $\cot A = \pm \left(4P - \frac{1}{16P}\right)$
- $\sqrt{\cos^2 A} = \cos A$ then $A \in Q_1, Q_4$
- Find m^2 and multiply

- Use formula $S_\infty = \frac{a}{1-r}$ for G.P
- If $(A+B) = 360^\circ$ or $(A+B) = 180^\circ$
 then $\cos^2 A = \cos^2 B$ and $(A+B) = 90^\circ$ then
 $(\cos^2 A + \cos^2 B) = 1$
- θ can be eliminated by substitution.
- Given $(a \sec \theta + b \tan \theta) = 1$
 $\Rightarrow (a \sec \theta - b \tan \theta) = 5$, eliminate θ
- Put $A = 30^\circ$ and $B = 45^\circ$ and verify
- Find $\tan \theta$ and simplify
- Put $A = 90^\circ$
- $m + n = a(\cos \alpha + \sin \alpha)^3$
 $m - n = a(\cos \alpha - \sin \alpha)^3$
- $a = \cos^2 \alpha - \sin^2 \alpha$
 $\Rightarrow 1 - a = 2 \sin^2 \alpha$ and $1 + a = 2 \cos^2 \alpha$
- In R.H.S. of given equation multiply and divide
 by $(1 + \cos x + \sin x)$
- $x - 1 = \tan \theta$, $y - 2 = \cot \theta$
 Multiply two equations
- $f(x) = \left(\sin^2 x + \frac{1}{\sin^2 x}\right) \geq 2$
 $\therefore \left(a + \frac{1}{a}\right) \geq 2$ for any real number $a > 0$
- for $a = 2$, $\cos \theta = \frac{5}{-3}$ not possible
- Put $\theta = 330^\circ$ substitution
- Put $n = 0$
- $\sin^2 x + \sin^4 x + \dots \infty = \frac{\sin^2 x}{1 - \sin^2 x} = \tan^2 x$
 $\therefore (e^{\log_e 2})^{\tan^2 x} = 8 \Rightarrow 2^{\tan^2 x} = 2^3 \Rightarrow \tan x = \sqrt{3}$
 $\therefore \operatorname{G.E} = \frac{1}{1 + \tan x} = \frac{1}{1 + \sqrt{3}} = \frac{\sqrt{3} - 1}{2}$

$$26. \left. \begin{array}{l} \cos x = \tan y \\ \tan z = \cot y \\ \cot z = \tan x \end{array} \right\} \text{multiplying we get } \cos x = \tan x$$

$$\Rightarrow \cos^2 x = \sin x$$

$$\Rightarrow \sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$27. \cos \theta + \sin \theta = p \Rightarrow p^2 = 1 + 2 \sin \theta \cos \theta \Rightarrow \sin \theta \cos \theta = \frac{p^2 - 1}{2}$$

also

$$p^3 = \cos^3 \theta + \sin^3 \theta + 3 \cos \theta \sin \theta (\cos \theta + \sin \theta)$$

$$\Rightarrow p^3 = q + 3 \left(\frac{p^2 - 1}{2} \right) p$$

$$\Rightarrow 2p^3 = 2q + 3p^3 - 3p \Rightarrow p^3 - 3p = -2q$$

$$28. \text{ If } 81^{\sin^2 x} = y \Rightarrow 81^{\cos^2 x} = \frac{81}{y}$$

$$\therefore y + \frac{81}{y} = 30 \Rightarrow y^2 - 30y + 81 = 0 \Rightarrow y = 27, 3$$

$$29. \because \operatorname{cosec} \alpha \in (-\infty, -1] \cup [1, \infty)$$

$$30. G.E = \frac{a \cos \theta - b \sin \theta}{a \sin \theta + b \cos \theta} = \frac{\pm \sqrt{b^2 - c^2 + a^2}}{c}$$



LEVEL - II (H.W)

1. If $\cos \theta = \frac{3}{5}$ and θ is not in the first quadrant,

$$\text{then } \frac{5 \tan(\pi + \theta) + 4 \cos(\pi + \theta)}{5 \sec(2\pi - \theta) - 4 \cot(2\pi + \theta)} =$$

$$1) \frac{4}{5} \quad 2) -\frac{4}{5} \quad 3) \frac{5}{4} \quad 4) -\frac{5}{4}$$

2. $f(x) = x^3 - 2x^2 + 3x - 5$

$$\Rightarrow f\left[\sin\left(\frac{5\pi}{2}\right)\right] + f\left[\sin\left(\frac{3\pi}{2}\right)\right] =$$

$$1) 10 \quad 2) -10 \quad 3) 14 \quad 4) -14$$

3. $\cos A, \sin A, \cot A$ are in GP

$$\text{then } \tan^6 A - \tan^2 A =$$

$$1) -1 \quad 2) 0 \quad 3) 1 \quad 4) 2$$

4. If $ABCD$ is a cyclic quadrilateral such that

$$12 \tan A - 5 = 0 \text{ and } 5 \cos B + 3 = 0, \text{ then}$$

$$\cos C \tan D =$$

$$1) \frac{-16}{13} \quad 2) \frac{16}{13} \quad 3) \frac{-13}{16} \quad 4) \frac{23}{16}$$

5. $x = \cos 1^\circ, y = \cos 1 \Rightarrow$

$$1) x = y \quad 2) x > y \quad 3) x < y \quad 4) 2x = y$$

6. $a = \sec 2^\circ, b = \sec 2 \Rightarrow$

$$1) a = b \quad 2) a < b \quad 3) b < a \quad 4) 2a = b$$

7. $\tan \theta = P - \frac{1}{4p} \Rightarrow \sec \theta - \tan \theta =$

$$1) 2p \text{ (or) } \frac{1}{2p} \quad 2) \frac{1}{2p} \text{ (or) } -2p$$

$$3) -\frac{1}{2p} \text{ (or) } 2p \quad 4) -\frac{1}{2p} \text{ (or) } -2p$$

8. If $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$, then A lies in the quadrants

$$1) \text{ I, II} \quad 2) \text{ II, III} \quad 3) \text{ I, III} \quad 4) \text{ I, IV}$$

9. $x = \tan \theta + \cot \theta, y = \cos \theta - \sin \theta \Rightarrow$

$$1) x = y \quad 2) \frac{1 - y^2}{2} = \frac{1}{x}$$

$$3) \frac{y^2 - 1}{2} = \frac{1}{x} \quad 4) \frac{1 + y^2}{2} = \frac{1}{x}$$

10. $1 + \cos x + \cos^2 x + \dots$ to $\infty = 4 + 2\sqrt{3}$ then $x =$

$$1) 30^\circ \quad 2) 60^\circ \quad 3) 45^\circ \quad 4) 90^\circ$$

11. $\sin^2 5^\circ + \sin^2 10^\circ + \dots + \sin^2 180^\circ =$

$$1) 18 \quad 2) 27 \quad 3) 1 \quad 4) 0$$

12. If $x = a \cos^3 \theta \sin^2 \theta, y = a \sin^3 \theta \cos^2 \theta$ and

$$\frac{(x^2 + y^2)^p}{(xy)^q}, (p, q \in \mathbb{N}) \text{ is independent of } \theta$$

then

- 1) $p + q = 6$ 2) $4p = 5q$
 3) $4q = 5p$ 4) $pq = 16$
13. $a = \sec \theta - \tan \theta$, $b = \operatorname{cosec} \theta + \cot \theta \Rightarrow a =$
 1) $\frac{b+1}{b-1}$ 2) $\frac{1+b}{1-b}$ 3) $\frac{b-1}{b+1}$ 4) $\frac{1-b}{1+b}$
14. $\cos A = a \cos B, \sin A = b \sin B \Rightarrow (b^2 - a^2) \sin^2 B =$
 1) $1 + a^2$ 2) $2 + a^2$ 3) $1 - a^2$ 4) $2 - a^2$
15. If $\frac{\sin x}{a} = \frac{\cos x}{b} = \frac{\tan x}{c} = k$ then
 $bc + \frac{1}{ck} + \frac{ak}{1+bk}$ is
 1) $k \left(a + \frac{1}{a} \right)$ 2) $\frac{1}{k} \left(a + \frac{1}{a} \right)$ 3) $\frac{1}{k^2}$ 4) $\frac{a}{k}$
16. If $m \cos^2 A + n \sin^2 A = p$, then $\cot^2 A =$
 1) $\frac{p+n}{m+p}$ 2) $\frac{p-n}{p-m}$ 3) $\frac{p-n}{m-p}$ 4) $\frac{n+1}{m+p}$
17. If $a \sin^3 x + b \cos^3 x = \sin x \cos x$
 and $a \sin x = b \cos x$ then $a^2 + b^2 =$
 1) 0 2) 1 3) 2 4) 3
18. If $\sin \theta + \cos \theta = a$, then $\sin^4 \theta + \cos^4 \theta =$
 1) $1 - \frac{1}{2}(a^2 + 1)^2$ 2) $1 - \frac{1}{2}(a^2 - 1)^2$
 3) $1 + \frac{1}{2}(a^2 + 1)^2$ 4) $1 + \frac{1}{2}(a^2 - 1)^2$
19. If $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{x}{\tan \theta - \sec \theta + 1}$ then $x =$
 1) 0 2) 2
 3) $\tan \theta - \sec \theta + 1$ 4) $\tan \theta + \sec \theta - 1$
20. If $\tan \theta + \sin \theta = m$, $\tan \theta - \sin \theta = n$, then
 $(m^2 - n^2)^2 =$
 1) $16mn$ 2) $4mn$ 3) $32mn$ 4) $8mn$
21. If $\sec \theta + \cos \theta = 2$ then $\sin^2 \theta + \tan^2 \theta =$
 1) 0 2) 1 3) 2 4) 4

22. Which of the following possible?
 1) $\sin \theta = 2$ 2) $\cos \theta = \frac{17}{5}$
 3) $\tan \theta = 2$ 4) $\sec \theta = \frac{4}{5}$
23. If $\theta = \frac{\pi}{21}$, then $\frac{\sin 23\theta - \sin 7\theta}{\sin 2\theta + \sin 14\theta} =$
 1) 0 2) 1 3) -1 4) 2
24. The value of $\tan \left(n\pi + \frac{\pi}{3} \right)$, $n \in I$ is
 1) $\sqrt{3}$ 2) $\frac{1}{\sqrt{3}}$ 3) 0 4) $\frac{1}{\sqrt{2}}$
25. If $e^{(1+\sin^2 x + \sin^4 x + \dots) \log 2} = 16$, then
 $\tan^2 x =$
 1) 1 2) 2 3) 3 4) 4
26. If $2 \sin x + 5 \cos y + 7 \sin z = 14$ then
 $7 \tan \frac{x}{2} + 4 \cos y - 6 \cos z =$
 1) 4 2) -3 3) 11 4) 5
27. If $\sin \theta$ and $\cos \theta$ are the roots of
 $px^2 + qx + r = 0$ then $q^2 - p^2 =$
 1) 0 2) $-2pr$ 3) $2qr$ 4) $2rp$
28. If $0 \leq x \leq \pi$, $4^{\sin^2 x} + 4^{\cos^2 x} = 5$, then $x =$
 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$

LEVEL-II(H.W)-KEY

- 01) 2 02) 4 03) 3 04) 1 05) 2
 06) 3 07) 2 08) 4 09) 2 10) 1
 11) 1 12) 2 13) 3 14) 3 15) 2
 16) 3 17) 2 18) 2 19) 4 20) 1
 21) 1 22) 3 23) 3 24) 1 25) 3
 26) 3 27) 4 28) 4

LEVEL-II(H.W)-HINTS

1. θ lies in fourth quadrant substitute the corresponding values.

2. $\sin \frac{3\pi}{2} = -1, \sin \frac{5\pi}{2} = 1$
- 3.. $\sin^2 A = \cos A \cdot \cot A \Rightarrow \sin^3 A = \cos^2 A$
 $\Rightarrow \tan^2 A = \cos A$ and simplify
4. $A + C = 180^\circ, B + D = 180^\circ$
 Given
 $\tan A = \frac{5}{12} > 0 \therefore 0 < A < \frac{\pi}{2} \Rightarrow \frac{\pi}{2} < C < \pi$
 $\cos B = \frac{-3}{5} < 0 \therefore \frac{\pi}{2} < B < \pi \Rightarrow 0 < D < \frac{\pi}{2}$
 $\therefore \cos C = \frac{-12}{13}; \tan D = \frac{4}{3}$
5. $1 > 1^0 \Rightarrow \cos 1 < \cos 1^0$
6. $2 = 2^c = 114^0$ (approx)
7. Use $(\sec^2 \theta - \tan^2 \theta) = 1$
8. $\sqrt{\cos^2 A} = \cos A$ then $A \in Q_1, Q_4$
9. Eliminate θ
10. Use formula $S_\infty = \frac{a}{1-r}$ for G.P
11. $A + B = 180^\circ \Rightarrow \sin^2 A = \sin^2 B$
 and $A + B = 90^\circ \Rightarrow \sin^2 A + \sin^2 B = 1$
12. $\frac{(x^2 + y^2)^p}{(xy)^q} = \frac{(a^2 \sin^4 \theta \cos^4 \theta)^p}{(a^2 \sin^5 \theta \cos^5 \theta)^q}$
 $= \frac{a^{2p} (\sin \theta \cos \theta)^{4p}}{a^{2q} (\sin \theta \cos \theta)^{5q}}$
 which is independent of θ , if $4p=5q$
13. Put $\theta = 45^\circ$ and verify the options
14. squaring and adding
15. $bc + \frac{1}{ck} + \frac{ak}{1+bk}$
 $= \frac{\sin x}{k^2} + \frac{\cos x(1 + \cos x) + \sin^2 x}{\sin x(1 + \cos x)}$

$$= \frac{a}{k} + \frac{1}{\sin x} = \frac{a}{k} + \frac{1}{ak} = \frac{1}{k} \left(a + \frac{1}{a} \right)$$

16. divided with $\sin^2 A$
 we get $\cot^2 A = \frac{p-n}{m-p}$
17. Put $\sin x = b$ and $\cos x = a$
 $\therefore a^2 + b^2 = \cos^2 x + \sin^2 x = 1$
18. Use $(a^4 + b^4) = (a^2 + b^2)^2 - 2a^2b^2$
19. In LHS divide the Num and Den by $\cos \theta$.
20. $2 \tan \theta = (m+n); \quad 2 \sin \theta = (m-n)$
 Eliminate θ and use $(\tan^2 \theta - \sin^2 \theta) = \tan^2 \theta \sin^2 \theta$
21. Put $\theta = 0$
22. Range of $\tan \theta$ is $\mathbb{R}$
23. Use $21\theta = \pi$
24. put $n = 0$
25. $(e^{\log 2})^{\frac{1}{1-\sin^2 x}} = 16 \Rightarrow 2^{\frac{1}{\cos^2 x}} = 2^4$
 $\Rightarrow \sec^2 x = 4 \Rightarrow \tan^2 x = 3$
26. put $x = \frac{\pi}{2}, y = 0, z = \frac{\pi}{2}$.
27. $\sin \theta + \cos \theta = \frac{-q}{p}, \sin \theta \cos \theta = \frac{r}{p}$
 now $(\sin \theta + \cos \theta)^2 = 1 + 2 \sin \theta \cos \theta$
28. $x = 90^\circ$, verify

LEVEL - III

1. $\tan^2 \alpha = 1 - p^2$, then $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha =$
 1) $(2+p^2)^{\frac{3}{2}}$ 2) $(1+p^2)^{\frac{3}{2}}$
 3) $(2-p^2)^{\frac{3}{2}}$ 4) $(1-p^2)^{\frac{3}{2}}$
2. $\frac{\pi}{2} < \alpha < \pi \Rightarrow \sqrt{\frac{1-\cos \alpha}{1+\cos \alpha}} + \sqrt{\frac{1+\cos \alpha}{1-\cos \alpha}} =$
 1) $2 \sec \alpha$ 2) $-2 \sec \alpha$

- 3) $2 \operatorname{cosec} \alpha$ 4) $-2 \operatorname{cosec} \alpha$
3. If $\sin \beta$ is the G.M. between $\sin \alpha$ and $\cos \alpha$ then $(\cos \alpha - \sin \alpha)^2 - 2 \cos^2 \beta =$
1) 0 2) 1 3) 2 4) -1
4. If $\sin^2 \theta = \frac{x^2 + y^2 + 1}{2x}$ then x must be
1) -3 2) -2 3) 1 4) 3
5. $\sin x + \sin^2 x + \sin^3 x = 1 \Rightarrow \cos^6 x - 4 \cos^4 x + 8 \cos^2 x =$
1) 4 2) 2 3) 1 4) 0
6. $\cos \theta + \cos^2 \theta = 1,$
 $a \sin^{12} \theta + b \sin^{10} \theta + c \sin^8 \theta + d \sin^6 \theta = 1 \Rightarrow \frac{b+c}{a+d} =$
1) 2 2) 3 3) 4 4) 6
7. $a \sin x = b \cos x = \frac{2c \tan x}{1 - \tan^2 x}$ and $(a^2 - b^2)^2 = kc^2(a^2 + b^2) \Rightarrow k =$
1) 1 2) 2 3) 3 4) 4
8. The greatest among $(\sin 1 + \cos 1), (\sqrt{\sin 1} + \sqrt{\cos 1}), (\sin 1 - \cos 1)$ and 1 is
1) $\sin 1 + \cos 1$ 2) $\sqrt{\sin 1} + \sqrt{\cos 1}$
3) $\sin 1 - \cos 1$ 4) 1
9. If $a \sin^2 x + b \cos^2 x = c, b \sin^2 y + a \cos^2 y = d$ and $a \tan x = b \tan y$ then $\frac{a^2}{b^2} =$
1) $\frac{(a-d)(c-a)}{(b-c)(d-b)}$ 2) $\frac{(b-c)(b-d)}{(a-c)(a-d)}$
3) $\frac{(b-c)(d-b)}{(a-d)(c-a)}$ 4) $\frac{(d-a)(c-a)}{(b-c)(d-b)}$
10. If $\tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma + \tan^2 \gamma \tan^2 \alpha + 2 \tan^2 \alpha \tan^2 \beta \tan^2 \gamma = 1$ then the value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$
1) 0 2) -1 3) 1 4) 2

11. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan \theta)^{\tan \theta},$
 $t_2 = (\tan \theta)^{\cot \theta}, t_3 = (\cot \theta)^{\tan \theta},$
 $t_4 = (\cot \theta)^{\cot \theta}$ then
1) $t_1 > t_2 > t_3 > t_4$ 2) $t_4 > t_3 > t_1 > t_2$
3) $t_3 > t_1 > t_2 > t_4$ 4) $t_2 > t_3 > t_1 > t_4$
12. Two arcs of same length of two different circles subtended angles of 25° and 75° at their centres respectively. Then the ratio of the radii of the circles is
1) 3 : 1 2) 1 : 3 3) 1 : 2 4) 2 : 1
13. If $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ where $x \in R, k \geq 1$ then $f_4(x) - f_6(x) =$ (MAINS-2014)
1) $\frac{1}{6}$ 2) $\frac{1}{3}$ 3) $\frac{1}{9}$ 4) $\frac{1}{12}$
14. The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as (MAINS-2013)
1) $\sin A \cos A + 1$ 2) $\sec A \operatorname{cosec} A + 1$
3) $\tan A + \cot A$ 4) $\sec A + \operatorname{cosec} A$
15. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$ then the acute angle 'R' is equal to [AIEEE-2012]
1) $\frac{\pi}{4}$ 2) $\frac{3\pi}{4}$ 3) $\frac{5\pi}{6}$ 4) $\frac{\pi}{6}$

LEVEL-III- KEY

- 01) 3 02) 3 03) 4 04) 3 05) 1
06) 2 07) 4 08) 2 09) 1 10) 3
11) 2 12) 1 13) 4 14) 2 15) 4

LEVEL-III- HINTS

1. $\sec \alpha \left[1 + \frac{\sin^3 \alpha \cos \alpha}{\cos^3 \alpha \sin \alpha} \right]$
 $= \sec \alpha [1 + \tan^2 \alpha] = \sec^3 \alpha$

$$= (\sec^2 \alpha)^{3/2} = (2 - P^2)^{3/2}$$

2. Simplifying we get $\frac{2}{\sqrt{\sin^2 \alpha}} = 2 \operatorname{cosec} \alpha$

$\therefore \sin \alpha$ is positive in II quadrant.

3. $\sin^2 \beta = \sin \alpha \cos \alpha$

4. $\sin^2 \theta \leq 1$

5. $\sin x(1 + \sin^2 x) = \cos^2 x$

Squaring both sides we get

$$(1 - \cos^2 x)(2 - \cos^2 x)^2 = \cos^4 x$$

Simplify further

6. $\cos \theta = 1 - \cos^2 \theta \Rightarrow \cos \theta = \sin^2 \theta$

$$\Rightarrow \cos^2 \theta = \sin^4 \theta \Rightarrow (\sin^4 \theta + \sin^2 \theta) = 1$$

Cubing both sides we get the answer

7. Take $a \sin x = b \cos x = c \tan 2x = l$

$$\Rightarrow \sin x = \frac{l}{a}, \cos x = \frac{l}{b}, \tan 2x = \frac{l}{c}$$

Elimination l we get

$$(a^2 - b^2)^2 = 4c^2(a^2 + b^2) \Rightarrow K = 4$$

8. $\sqrt{\sin 1} > \sin 1 > \sin^2 1$

$$\sqrt{\cos 1} > \cos 1 > \cos^2 1$$

$$\text{Adding } (\sqrt{\sin 1} + \sqrt{\cos 1}) > (\sin 1 + \cos 1)$$

$$> (\sin^2 1 + \cos^2 1)$$

$$\Rightarrow (\sqrt{\sin 1} + \sqrt{\cos 1}) > (\sin 1 + \cos 1) > 1$$

9. $a \sin^2 x + b \cos^2 x = c$

$$\Rightarrow a \tan^2 x + b = c \sec^2 x$$

$$\Rightarrow (a - c) \tan^2 x = c - b \Rightarrow \tan^2 x = \frac{c - b}{a - c}$$

$$\text{similarly } \tan^2 y = \frac{d - a}{b - d}$$

$$\text{but } a \tan x = b \tan y \Rightarrow \frac{a^2}{b^2} = \frac{\tan^2 y}{\tan^2 x}$$

10. Let $\tan^2 \alpha = x, \tan^2 \beta = y, \tan^2 \gamma = z$

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \sum \frac{\tan^2 \alpha}{1 + \tan^2 \alpha} = \sum \frac{x}{1 + x}$$

$$= \frac{(x + y + z) + (xy + yz + zx + 2xyz) + xy + yz + zx + xyz}{(1 + x)(1 + y)(1 + z)}$$

$$= \frac{1 + x + y + z + xy + yz + zx + xyz}{(1 + x)(1 + y)(1 + z)} = 1$$

11. In $\left(0, \frac{\pi}{4}\right), \tan \theta < 1$ and $\cot \theta > 1$

$$(\cot \theta)^{\cot \theta} > (\cot \theta)^{\tan \theta} > (\tan \theta)^{\tan \theta} > (\tan \theta)^{\cot \theta}$$

$$\therefore t_4 > t_3 > t_1 > t_2$$

12. Length of arc, $l = r\theta$

$$\therefore \frac{r_1}{r_2} = \frac{l/\theta_1}{l/\theta_2} = \frac{\theta_2}{\theta_1} = \frac{75}{25} = \frac{3}{1}$$

13. $f_4(x) = 1 - 2 \sin^2 \theta \cos^2 \theta$ and

$$f_6(x) = 1 - 3 \sin^2 \theta \cos^2 \theta$$

14. $\frac{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}}{1 - \frac{\cos A}{\sin A}} = \frac{\frac{\sin^2 A}{\cos A(\sin A - \cos A)} + \frac{\cos^2 A}{\sin A(\cos A - \sin A)}}{1 - \frac{\cos A}{\sin A}}$

$$= \frac{\sin^3 A - \cos^3 A}{\sin A \cos A (\sin A - \cos A)} = \frac{\sin^2 A + \sin A \cos A + \cos^2 A}{\sin A \cos A}$$

$$= \frac{1 + \sin A \cos A}{\sin A \cos A} = \sec A \operatorname{cosec} A + 1$$

15. Squaring both the equations and adding

LEVEL - IV

1. **Assertion A:** In a right angled triangle

$$\sin^2 A + \sin^2 B + \sec^2 C = 2.$$

Reason R: If α, β are complementary angles

then $\sin^2 \alpha + \sin^2 \beta = 1$

1) A, R are true and R is the correct explanation of A

- 2) A, R are true and R is not the correct explanation of A
 3) A is true, R is false
 4) A is false, R is true

2. Observe the following lists. Let

$$\frac{\sin x}{a} = \frac{\cos x}{b} = \frac{\tan x}{c} = k.$$

List - I

List-II

- | | |
|-------------------------------------|------------------------|
| 1) bc | a) $\frac{1}{b^2 k^4}$ |
| 2) $a^2 + b^2$ | b) $\frac{1}{ak}$ |
| 3) $\frac{1}{ck} + \frac{ak}{1+bk}$ | c) $\frac{a}{k}$ |
| 4) $a^2 + b^2 + c^2$ | d) $\frac{1}{k^2}$ |

The correct match for List-I from List-II is

- 1) 1 - c, 2 - d, 3 - b, 4 - a
 2) 1 - d, 2 - a, 3 - c, 4 - b
 3) 1 - a, 2 - b, 3 - d, 4 - c
 4) 1 - b, 2 - c, 3 - a, 4 - d

3. Observe the following lists.

List-I

List-II

- | | |
|---|-------------------|
| I. $\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4}$ | a) $\frac{15}{2}$ |
| II. $\sin^2 \frac{11\pi}{6} + \cos^2 \frac{7\pi}{6} - \tan^2 \frac{5\pi}{4}$ | b) $-\frac{1}{2}$ |
| III. $\sin^2 \frac{3\pi}{4} + \sec^2 \frac{5\pi}{3} + \tan^2 \frac{2\pi}{3}$ | c) 0 |
| IV. $\cos^2 \frac{2\pi}{3} + \sin^2 \frac{5\pi}{3} - \frac{3}{2} \tan^2 \frac{3\pi}{4}$ | d) $\frac{1}{2}$ |

The correct match for List-I from List-II is

- 1) I-d, II-c, III-a, IV-b 2) I-a, II-b, III-c, IV-d
 3) I-b, II-a, III-d, IV-c 4) I-c, II-d, III-b, IV-a

4. **Satement I:** The number of values of θ satisfying $\sec \theta + \cos \theta = 1$ is 2.

Statement II: If $7 \sin^2 \alpha + 3 \cos^2 \alpha = 4$, then $\tan \alpha = \sqrt{3}$.

Which of the above statements is correct ?

- 1) only I is true 2) only II is true
 3) Both I and II are true
 4) Neither I nor II is true

5. If $\alpha = \cos 10^\circ - \sin 10^\circ$,

$$\beta = \cos 45^\circ - \sin 45^\circ,$$

$\gamma = \cos 70^\circ - \sin 70^\circ$ then the descending order of α, β, γ is

- 1) α, β, γ 2) γ, β, α 3) α, γ, β 4) β, α, γ

6. Match the following:

List - I

List - II

- 1) $3 \tan x + 27 \cot x \geq$ a) 24

$$(x \in Q_1)$$

- 2) $5 \sec^2 x + 125 \cos^2 x \geq$ b) 18

- 3) $16 \cos^2 x + 9 \sin^2 x \geq$ c) 50

- 1) 1-a, 2-b, 3-c 2) 1-c, 2-a, 3-b

- 3) 1-b, 2-c, 3-a 4) 1-c, 2-b, 3-a

7. **A = tan1, B = tan2, C = tan3**, then the descending order of A, B, C is

- 1) A, B, C 2) C, B, A 3) A, C, B 4) B, C, A

LEVEL-IV-KEY

- 01) 4 02) 1 03) 1 04) 4 05) 1
 06) 3 07) 3

LEVEL-IV-HINTS

- In a right angled triangle
 $\sin^2 A + \sin^2 B + \sec^2 C \neq 2$
- Simplify
- Simplify
- I. $(\sec \theta + \cos \theta) = 1$ is not possible for $\theta \in R$
 II. Divide given equation by $\cos^2 \alpha$
- α is positive, β is zero,
 γ is negative
- $A.M \geq G.M$
 $a \tan x + b \cot x \geq 2\sqrt{ab}$
- $1 = 57^\circ$
 $2 = 114^\circ$
 $3 = 171^\circ$ (Aprox)